

AI RISK MANAGER TRACK

AI Risk Investigation & Early Warning Agent

"From suspicious transactions to explainable risk investigations."



Transaction



Risk Detection



Early Warning



Investigation



Response



Audit

Fraud Doesn't Always Start With One Suspicious Transaction

PROBLEM STATEMENT

TRADITIONAL MONITORING FOCUS



- ✗ Attackers distribute activity across cards, users, devices & IPs to bypass rules.
- ✗ Individual events appear harmless when evaluated in isolation.
- ✗ Coordinated behavior emerges long before a traditional threshold alert triggers.
- ✗ Investigators receive isolated scores without connected entity evidence.

"The real challenge is detecting when small risk signals begin forming a larger pattern."

EMERGING COORDINATED RISK CLUSTER



Multiple low-scoring events converge through shared identity/device graphs.

From Transaction Detection to Risk Investigation

SYSTEM SOLUTION



Transaction Data



Risk Detectors



Risk Scoring



Early Warning



Relationship Graph



AI Agent

Recommended
Response



Audit Trail

▼ RISK DETECTORS LAYER



Card Testing



Account Takeover



Device / IP Coordinated Risk

Specialized statistical & rule engines flag micro-anomalies across transaction velocity and entity metadata.

🔗 RELATIONSHIP GRAPH MAPPING

User ↔ Device

Device ↔ IP

User ↔ Payment ID

Merchant ↔ Transactions

Constructs real-time entity links to uncover shared infrastructure among seemingly independent accounts.

"Our system combines individual risk signals with connected-entity intelligence to identify emerging coordinated risk."

How the AI Risk Investigation Works

MULTI-LAYER WORKFLOW

STAGE 01

1. DETECT

Identify suspicious behavioural and transaction signals across payment streams using velocity & temporal anomalies.

STAGE 02

2. CONNECT

Link users, devices, IP addresses, payment identifiers, and merchants into a dynamic risk network graph.

STAGE 03

3. INVESTIGATE

Combine risk signals and generate an evidence-based, human-readable investigation explanation using LLM reasoning.

STAGE 04

4. RESPOND

Recommend a bounded action such as verification, human review, step-up auth, or risk escalation.

 Evidence



 Explanation



 Decision



Audit Trail

 **Responsible AI Guardrail:** AI recommends; consequential actions remain controlled by deterministic rules and human risk investigators.

Investigator Console — Live Demo

● STREAMLITAPP

PRODUCT INTERFACE

TOTAL TRANSACTIONS

1,248

HIGH RISK

14

CRITICAL

3

EARLY WARNINGS

8

AVG RISK SCORE

34.2

📈 RISK EVOLUTION

ACTIVE RING

- Highest-Risk Events: **Ring #04 (Card Testing)**
- Connected Risk Clusters: **2 Active Rings**
- Temporal Velocity: **+180% surge in 10 mins**

DEMO MODE

Ready for live dashboard demonstration

⚙️ SELECTED DEMO TRANSACTION

CRITICAL

Transaction ID

135df81a-a77a-4985-a637-ee84708af87c

Risk Score / Level

100 / CRITICAL

Warning Status

CRITICAL_WARNING

Recommended Action

HOLD_FOR_VERIFICATION

🔍 AI Investigation Explanation

"Evidence from the actual dashboard will be demonstrated live."

Technical Architecture

SYSTEM DESIGN



TECHNOLOGY STACK

Python 3.11

Pandas / NumPy

FastAPI

Streamlit

AI Investigation Layer

Graph-based Risk Analysis

Note: Built as a working prototype for the buildathon. Current build uses synthetic transaction data for model validation prior to production integration.

Why This Approach Matters

DIFFERENTIATION & VALUE

Traditional Approach

Transaction

└─▶ Risk Score

└─▶ Alert (Isolated)

- Evaluates transactions independently
- High false-positive rate on surges
- Black-box scores without reasoning

Our AI Agent Approach

Transaction → Risk Signals → Emerging Pattern → Connected Entities
→ AI Investigation → Evidence Explanation → Bounded Response → Audit

✓ Early-warning intelligence

✓ Coordinated-risk visibility

✓ Explainable investigation

✓ Bounded AI responses

✓ Full Auditability

✓ Investigator workflow

**"We are not building another transaction risk score.
We are building an AI investigation layer that helps risk teams understand emerging threats before they become larger incidents."**

Limitations & Scope Discipline: Current prototype uses synthetic transaction data and requires further validation before production deployment. Strictly defense-only.